Module 21: ASA Firewall Configuration

Networking Security v1.0
(NETSEC)

Module Objectives

Module Title: ASA Firewall Configuration

Module Objective: Implement an ASA firewall configuration.

Topic Title	Topic Objective
Basic ASA Firewall Configuration	Explain how to configure an ASA-5506-X with FirePOWER Services.
Configure Management Settings and Services	Configure management settings and services on an ASA 5506-X firewall.
Object Groups	Explain how to configure object groups on an ASA.
ASA ACLs	Use the correct commands to configure access lists with object groups on an ASA.
NAT Services on an ASA	Use the correct commands to configure an ASA to provide NAT services.
AAA	Use correct commands to configure access control using the local database and AAA server.
Service Policies on an ASA	Configure service policies on an ASA
Introduction to ASDM (Optional)	Note: This is an optional topic that is not assessed.

21.1 Basic ASA Firewall Configuration

Basic ASA Settings

The ASA command line interface (CLI) is a proprietary OS, which has a similar look and feel to the router IOS.

For example, the ASA CLI contains command prompts similar to that of a Cisco IOS router.

- Abbreviation of commands and keywords
- Use of the Tab key to complete a partial command
- Use of the help key (**?**) after a command to view additional syntax

Basic ASA Settings (Cont.)

The table contrasts common IOS router and ASA commands.

IOS Router Command	Equivalent ASA Command
enable secret *password*	enable password *password*
line vty 0 4 password *password* login	passwd *password*
ip route	route *if_name*
show ip interface brief	show interface ip brief
show ip route	show route
show ip nat translations	show xlate
copy running-config startup-config	write [memory]
erase startup-config	write erase

Basic ASA Settings (Cont.)

ASA CLI commands can be executed regardless of the current configuration mode prompt.

```
ciscoasa# conf t
ciscoasa(config)# show password encryption
Password Encryption: Disabled
Master key hash: Not set(saved)
ciscoasa(config)#
```

ASA CLI commands can be executed regardless of the current configuration mode prompt. The IOS do command is not required nor recognized.

ASA Default Configuration

The ASA 5506-X with FirePOWER Services ships with a default configuration that, in most instances, is sufficient for a basic SOHO deployment. Use the **configure factory-default** global configuration mode command to restore the factory default configuration.

The default hostname is **ciscoasa**. By default, the privileged EXEC and console line passwords are not configured.

These settings can be changed by:

- Manually using the CLI
- Interactively using the CLI Setup Initialization wizard
- Using the ASDM Startup wizard

```
hostname ciscoasa
enable password 8Ry2YjIyt7RRXU24 encrypted
names
no mac-address auto

!
interface GigabitEthernet1/1
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet1/2
 shutdown
 no nameif
 no security-level
 no ip address
!
<output omitted>
```

ASA Interactive Setup Initialization Wizard

The wizard is displayed when there is no startup configuration, and prompts “**Pre-configure Firewall now through interactive prompts [yes]?**” To cancel and display the ASA default user EXEC mode prompt, enter **no**. Otherwise, enter **yes** or simply press **Enter** to accept the default [yes]. This initiates the wizard and the ASA interactively guides an administrator to configure the default settings.

```
Pre-configure Firewall now through interactive prompts [yes]? <Enter>
Firewall Mode [Routed]: <Enter>
Enable password [<use current password>]: cisco
Allow password recovery [yes]? <Enter>
Clock (UTC):
  Year [2021]:
  Month [Feb]:
  Day [9]:
  Time [11:21:11]:
Management IP address: 192.168.1.1
Management network mask: 255.255.255.0
Host name: NETSEC-ASA
Domain name: netsec.com
IP address of host running Device Manager: 192.168.1.100

The following configuration will be used:
Enable password: cisco
Allow password recovery: yes
Clock (UTC): 11:21:11 Feb 9 2021
Firewall Mode: Routed
Management IP address: 192.168.1.1
Management network mask: 255.255.255.0
Host name: NETSEC-ASA
Domain name: netsec.com
IP address of host running Device Manager: 192.168.1.100

Use this configuration and save to flash? [yes]<Enter>
```

21.2 Configure Management Settings and Services

Enter Global Configuration Mode

The default ASA user prompt of **ciscoasa>** is displayed when an ASA configuration is erased, the device is rebooted, and the user does not use the interactive setup wizard.

To enter privileged EXEC mode, use the **enable** user EXEC mode command. Initially, an ASA does not have a password configured; therefore, when prompted, leave the enable password prompt blank and press **Enter**.

The ASA date and time should be set either manually or by using Network Time Protocol (NTP). To set the date and time, use the **clock set** privileged EXEC **command.**

Enter global configuration mode using the **configure terminal** privileged EXEC command.

Configure Basic Settings

An ASA must be configured with basic management settings. The table displays the commands to accomplish this task.

ASA Command	Description
hostname *name*	• Specifies a hostname up to 63 characters. • A hostname must start and end with a letter or digit, and have as interior characters only letters, digits, or a hyphen.
domain-name *name*	Sets the default domain name.
enable password *password*	• Sets the enable password for privileged EXEC mode. • Sets the password as a case-sensitive string of 3 to 32 alphanumeric and special characters (not including a question mark or a space).
banner motd *message*	Provides legal notification and configures the system to display a message-of-the-day banner when connecting to the ASA.
key config-key password-encryption [*new-pass* [*old-pass*]]	• Sets the passphrase between 8 and 128 character long. • Used to generate the encryption key.
password encryption aes	Enables password encryption and encrypts all user passwords.

Configure Basic Settings (Cont.)

To configure a banner with several lines, the **banner motd** must be entered multiple times.

The privileged EXEC password is automatically encrypted using MD5. However, stronger encryption using AES should be enabled. To do so, a master passphrase must be configured, and AES encryption must be enabled.

To change the master passphrase, use the **key config-key password-encryption** *password* command. To determine if password encryption is enabled, use the **show password encryption** command.

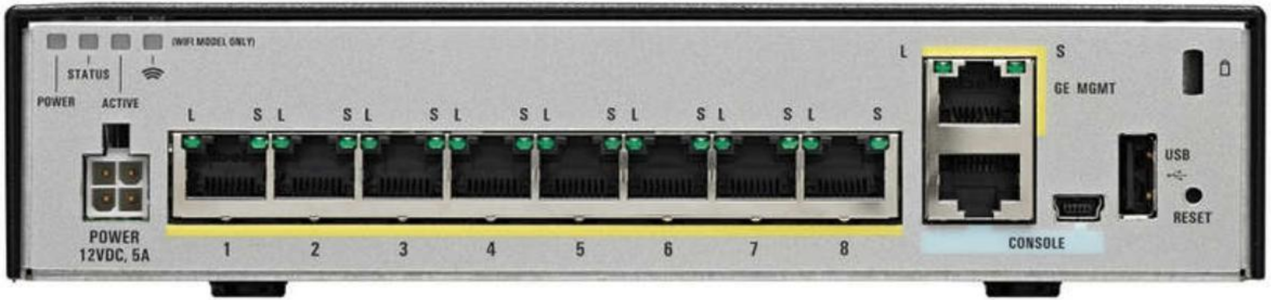
```
NETSEC-ASA> enable
Password: *********
NETSEC-ASA# show password encryption
Password Encryption: Disabled
Master key hash: Not set(saved)
NETSEC-ASA#
NETSEC-ASA# configure terminal
NETSEC-ASA(config)# key config-key password-encryption cisco123
NETSEC-ASA(config)# password encryption aes
NETSEC-ASA(config)# exit
NETSEC-ASA#
NETSEC-ASA# show password encryption
Password Encryption: Enabled
Master key hash: 0x45ebef8e 0x77a0f287 0x90247f80 0x2a184246 0xe85cbcc4(not saved)
NETSEC-ASA# write
Building configuration...
Cryptochecksum: c2cb4c42 66ed8038 c81a3d7f c5df996e

6781 bytes copied in 0.260 secs
[OK]
NETSEC-ASA#
```


Configure Interfaces

The ASA-5506-X has eight Gigabit Ethernet interfaces that can be configured to carry traffic from different networks. The G1/1 interface is used by convention as the outside interface and is set to receive its IP address over DHCP by default.

The remaining interfaces, G1/2-G1/8, can be assigned to inside networks or DMZs. In addition, a Gigabit Ethernet port is dedicated to in-band management. It is designated as Management1/1. There are also RJ45 and USB console connections for out-of-band management.



Configure Interfaces (Cont.)

The IP address of an interface can be configured using one of the following options:

- **Manually -** Commonly used to assign an IP address and mask to the interface.
- **DHCP -** Used when an interface is connecting to an upstream device providing DHCP services.
- **PPPoE -** Used when an interface is connecting to an upstream DSL device providing point-to-point connectivity over Ethernet services

The table lists the commands to configure an IP address on an interface.

To Configure	ASA Command	Description
Manually	**ip address** *ip-address netmask*	Assigns an IP address to the interface.
Using DHCP	**ip address dhcp**	The interface will request an IP address configuration from the upstream device.
	ip address dhcp setroute	Used to have the interface request and install a default route to the upstream device.
Using PPPOE	**ip address pppoe**	Interface configuration mode command that requests an IP address from the upstream device.
	ip address pppoe setroute	Same command but it also requests and installs a default route to the upstream device.

Configure Interfaces (Cont.)

Each interface must have a security level from 0 (lowest) to 100 (highest).

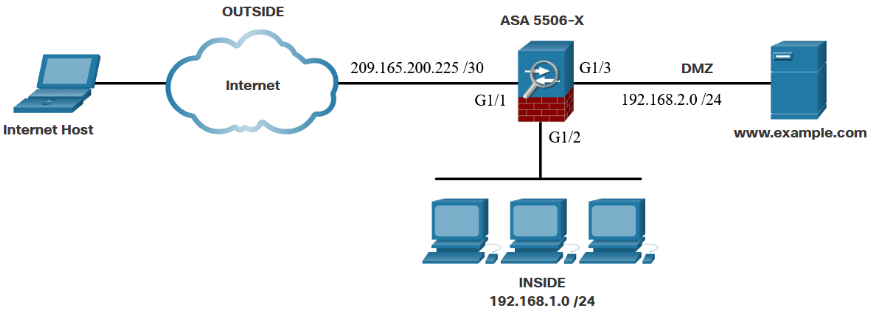
The commands below are used to configure basic interface parameters.

Verify the interface addressing and status with the **show interface ip brief** command as shown below. Note that the **show** command does not need to be entered in User EXEC mode.

ASA Command	Description
nameif *if_name*	• Names the interface using a text string of up to 48 characters. • The name is not case-sensitive. • You can change the name by re-entering this command with a new value. • Do not enter the **no** form, because that command causes all commands that refer to that name to be deleted.
security-level *value*	Sets the security level, where number is an integer between 0 (lowest) and 100 (highest).
no shutdown	Activate the interface.

Configure Interfaces (Cont.)

Interface configuration example




```
NETSEC-ASA(config)# interface g1/1
NETSEC-ASA(config-if)# nameif OUTSIDE
INFO: Security level for "OUTSIDE" set to 0 by default.
NETSEC-ASA(config-if)# security-level 0
NETSEC-ASA(config-if)# ip address 209.165.200.225 255.255.255.252
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config-if)# interface g1/2
NETSEC-ASA(config-if)# nameif INSIDE
INFO: Security level for "INSIDE" set to 100 by default.
NETSEC-ASA(config-if)# security-level 100
NETSEC-ASA(config-if)# ip address 192.168.1.1 255.255.255.0
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config-if)# interface g1/3
NETSEC-ASA(config-if)# nameif DMZ
INFO: Security level for "DMZ" set to 0 by default.
NETSEC-ASA(config-if)# security-level 50
NETSEC-ASA(config-if)# ip address 192.168.2.1 255.255.255.0
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config-if)# exit
NETSEC-ASA(config)#
```

Configure a Default Static Route

Use the **route** *interface-name* **0.0.0.0 0.0.0.0** *next-hop-ip-address* command to configure a default route.

```
NETSEC-ASA(config)# route OUTSIDE 0.0.0.0 0.0.0.0 209.165.200.226
NETSEC-ASA(config)#
NETSEC-ASA(config)# show route | begin Gateway
Gateway of last resort is 209.165.200.226 to network 0.0.0.0

S*       0.0.0.0 0.0.0.0 [1/0] via 209.165.200.226, OUTSIDE
C        192.168.1.0 255.255.255.0 is directly connected, INSIDE
L        192.168.1.1 255.255.255.255 is directly connected, INSIDE
C        192.168.2.0 255.255.255.0 is directly connected, DMZ
L        192.168.2.1 255.255.255.255 is directly connected, DMZ
C        209.165.200.224 255.255.255.252 is directly connected, OUTSIDE
L        209.165.200.225 255.255.255.255 is directly connected, OUTSIDE

NETSEC-ASA(config)#
```

Configure Remote Access Services

Telnet or SSH is required to manage the ASA 5506-X remotely, using the CLI. To enable the Telnet service, use the commands listed in the table.

ASA Command	Description
{**passwd** | **password**} *password*	Sets the login password up to 80 characters in length for Telnet.
telnet { *ipv4_address mask* | *ipv6_address/prefix* } *if_name*	• Identifies which inside host or network can Telnet to the ASA interface. • Use the clear configure telnet command to remove the Telnet connection
telnet timeout *minutes*	• By default, Telnet sessions left idle for five minutes are closed by the ASA. • The command alters the default exec timeout of five minutes.
aaa authentication telnet console LOCAL	• Configures Telnet to refer to the local database for authentication. • The LOCAL keyword is case sensitive and is a predefined server tag.
clear configure telnet	Removes the Telnet connection from the configuration.

```
NETSEC-ASA(config)# password cisco
NETSEC-ASA(config)# telnet 192.168.1.3 255.255.255.255 INSIDE
NETSEC-ASA(config)# telnet timeout 3
NETSEC-ASA(config)#
NETSEC-ASA(config)# show run telnet
telnet 192.168.1.3 255.255.255.255 INSIDE
telnet timeout 3
NETSEC-ASA(config)#
```

Configure Remote Access Services (Cont.)

To enable SSH, use the commands that are listed in the table. Example configuration is on the next slide

ASA Command	Description
username *name* **password** *password*	Creates a local database entry.
aaa authentication ssh console LOCAL	• Configures SSH to refer to the local database for authentication. • The **LOCAL** keyword is case sensitive and is a predefined server tag.
crypto key generate rsa modulus *modulus_size*	• Generates the RSA key required for SSH encryption. • The *modulus_size* (in bits) can be 512, 768, 1024, or 2048. • A value of 2048 is recommended.
ssh { *ip_address mask* | *ipv6_address/prefix* } **if_name**	• Identifies which inside host or network can SSH to the ASA interface. • Multiple commands can be in the configuration. • If the *if_name* is not specified, SSH is enabled on all interfaces except the outside interface. • Use the clear configure ssh command to remove the SSH connection.
ssh version *version_number*	• (Optional) By default, the ASA allows both SSH Version 1 (less secure) and Version 2 (more secure). • Enter this command in order to restrict the connections to a specific version.
ssh timeout *minutes*	Alters the default exec timeout of five minutes.
clear configure ssh	Removes the SSH connection from the configuration.

Configure Remote Access Services (Cont.)

```
NETSEC-ASA(config)# username ADMIN password class
NETSEC-ASA(config)# aaa authentication ssh console LOCAL
NETSEC-ASA(config)# crypto key generate rsa modulus 2048
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

Do you really want to replace them? [yes/no]: y
Keypair generation process begin. Please wait...
NETSEC-ASA(config)# ssh 192.168.1.3 255.255.255.255 INSIDE
NETSEC-ASA(config)# ssh 192.168.1.4 255.255.255.255 INSIDE
NETSEC-ASA(config)# ssh 172.16.1.3 255.255.255.255 OUTSIDE
NETSEC-ASA(config)# ssh version 2
NETSEC-ASA(config)# show ssh
Timeout: 5 minutes
Version allowed: 2
Cipher encryption algorithms enabled:    aes256-ctr   aes256-cbc   aes192-ctraes192-cbc
aes128-ctr   aes128-cbc
Cipher integrity  algorithms enabled:    hmac-sha2-256

Hosts allowed to ssh into the system:
172.16.1.3 255.255.255.255 OUTSIDE
192.168.1.3 255.255.255.255 INSIDE
192.168.1.4 255.255.255.255 INSIDE
NETSEC-ASA(config)#
```

Optional Lab - Configure ASA Basic Settings Using the CLI

In this lab, you will complete the following objectives:

- Part 1: Configure Basic Device Settings
- Part 2: Access the ASA Console and Use CLI Setup Mode to Configure Basic Settings
- Part 3: Configure Basic ASA Settings and Interface Security Levels

Configure Network Time Protocol Services

Network Time Protocol (NTP) services can be enabled on an ASA to obtain the date and time from an NTP server. To enable NTP, use the global configuration mode commands listed in the table. To verify the NTP configuration and status, use the **show ntp status** and **show ntp associations** commands

ASA Command	Description
ntp authenticate	Enables authentication with an NTP server.
ntp trusted-key *key_id*	Specifies an authentication key ID to be a trusted key, which is required for authentication with an NTP server.
ntp authentication-key *key_id* **md5** *key*	Sets a key to authenticate with an NTP server.
ntp server *ip_address* [**key** *key_id*]	Identifies an NTP server.

```
NETSEC-ASA(config)# ntp authenticate
NETSEC-ASA(config)# ntp trusted-key 1
NETSEC-ASA(config)# ntp authentication-key 1 sha-256 cisco123
NETSEC-ASA(config)# ntp server 192.168.1.254
NETSEC-ASA(config)#
```

Configure DHCP Services

An ASA can be configured to be a DHCP server to provide IP addresses and DHCP-related information to hosts. To enable an ASA as a DHCP server and provide DHCP services to hosts, use the commands listed in the table. Example on the next slide

ASA Command	Description
dhcpd address *IP_address1* [- IP_address2] *if_name*	• Creates a DHCP address pool in *which IP_address1* is the start of the pool and IP_address2 is the end of the pool, separated by a hyphen. • The address pool must be on the same subnet as the ASA interface.
dhcpd dns *dns1* [*dns2*]	(Optional) Specifies the IP address(es) of the DNS server(s).
dhcpd lease *lease_length*	• (Optional) Changes the lease length granted to the client which is the amount of time in seconds that the client can use its allocated IP address before the lease expires. • The *lease_length* defaults to 3600 seconds (1 hour) but can be a value from 0 to 1,048,575 seconds.
dhcpd domain *domain_name*	(Optional) Specifies the domain name assigned to the client.
dhcpd enable *if_name*	Enables the DHCP server service (daemon) on the interface (typically the inside interface) of the ASA.

Configure DHCP Services (Cont.)

```
NETSEC-ASA(config)# dhcpd address 10.0.0.1-10.0.1.255 INSIDE
Warning, DHCP pool range is limited to 256 addresses, set address range as: 10.0.0.1-10.0.1.0
Address range subnet 10.0.0.1 or 10.0.1.0 is not the same as INSIDE interface subnet
192.168.1.1
NETSEC-ASA(config)# dhcpd address 192.168.1.10-192.168.1.250 INSIDE
NETSEC-ASA(config)# dhcpd lease 1800
NETSEC-ASA(config)#
```

21.3 Object Groups

Introduction to Objects and Object Groups

Objects are reusable components for use in configurations. Objects can be defined and used in Cisco ASA configurations in the place of inline IP addresses, services, names, and so on.

The ASA supports objects and object groups, as shown in the output in the following output from the help facility:

```
NETSEC-ASA(config)# object ?

configure mode commands/options:
  network  Specifies a host, subnet or range IP addresses
  service  Specifies a protocol/port
NETSEC-ASA(config)#
NETSEC-ASA(config)# object-group ?

configure mode commands/options:
  icmp-type  Specifies a group of ICMP types, such as echo
  network    Specifies a group of host or subnet IP addresses
  protocol   Specifies a group of protocols, such as TCP, etc
  security   Specifies identity attributes such as security-group
  service    Specifies a group of TCP/UDP ports/services
  user       Specifies single user, local or import user group
NETSEC-ASA(config)#
```

Configure Network Objects

To create a network object, use the **object network** *object-name* global configuration mode command. The prompt changes to network object configuration mode.

Network objects can consist of the following:

- **host** - a host address
- **fqdn** - a fully-qualified domain name
- **range** - a range of IP addresses
- **subnet** - an entire IP network or subnet

Configure Network Objects (Cont.)

Commands available in network object configuration mode are shown in the table.

ASA Command	Description
attribute *attribute-agent attribute-type attribute-value*	Defined and used to filter traffic associated with one or more virtual machines.
description	Enter a description of the object up to 200 characters in length.
fqdn	A fully-qualified domain name such as the name of a host, such as www.example.com. Specify v4 to limit the address to IPv4, and v6 for IPv6. If you do not specify an address type, IPv4 is assumed.
host *ip-address*	The IPv4 or IPv6 address of a single host.
range *start_add end_add*	A range of addresses. You can specify IPv4 or IPv6 ranges. Do not include masks or prefixes.
subnet {ipv4_add ipv4_mask | ipv6_add/ipv6_prefix}	Assigns a network subnet to the named object.

Configure Network Objects (Cont.)

The example displays a sample network object configuration. Notice that the configuration of **range** overwrites the previous configuration of **host**.

```
NetSec-ASA(config)# object network EXAMPLE-1
NetSec-ASA(config-network-object)# host 192.168.1.3
NetSec-ASA(config-network-object)# exit
NetSec-ASA(config)# show run object
object network EXAMPLE-1
 host 192.168.1.3
NetSec-ASA(config)# object network EXAMPLE-1
NetSec-ASA(config-network-object)# range 192.168.1.10 192.168.1.20
NetSec-ASA(config-network-object)# exit
NetSec-ASA(config)# show run object
object network EXAMPLE-1
 range 192.168.1.10 192.168.1.20
NetSec-ASA(config)#
```

Configure Service Objects

The table provides an overview of common service options available. Optional keywords are used to identify source port or destination port, or both. Operators such as **eq** (equal), **neq** (not equal), **lt** (less than), **gt** (greater than), and **range**, support configuring a port for a given protocol. If no operator is specified, the default operator is **eq**.

Use the **no** form of the command to remove a service object. To erase all service objects, use the **clear config object service** command.

ASA Command	Description
service *protocol*	Specifies an IP protocol name or number.
service tcp [**source** *operator port*] [**destination** *operator port*]	Specifies that the service object is for the TCP protocol.
service udp [**source** *operator port*] [**destination** *operator port*]	Specifies that the service object is for the UDP protocol.
service icmp *[icmp-type [icmp_code]]*	Specifies that the service object is for the ICMP protocol.
service icmp6 *[icmp-type [icmp_code]]*	Specifies that the service object is for the ICMPv6 protocol.

Configure Service Objects (Cont.)

A service object name can only be associated with one protocol and port (or ports), as shown in the **show run object service** output in this example.

```
NETSEC-ASA(config)# object service SERV-1
NETSEC-ASA(config-service-object)# service tcp destination eq ftp
NETSEC-ASA(config-service-object)# service tcp destination eq www
NETSEC-ASA(config-service-object)# exit
NETSEC-ASA(config)# show run object service
object service SERV-1
 service tcp destination eq www
NETSEC-ASA(config)#
```

Object Groups

Objects can be grouped together to create an object group. By grouping like objects together, an object group can be used in an access control entry (ACE) instead of having to enter an ACE for each object separately.

The following guidelines and limitations apply to object groups:

- Objects and object groups share the same name space.
- Object groups must have unique names.
- An object group cannot be removed or emptied if it is used in a command.
- The ASA does not support IPv6 nested object groups.

Object Groups (Cont.)

There are five types of object groups.

- **Network** - A network-based object group specifies a list of IP host, subnet, or network addresses.
- **User** - Locally created, as well as imported Active Directory user groups can be defined for use in features that support the identity firewall.
- **Service** - A service-based object group is used to group TCP, UDP, or TCP and UDP ports into an object. The ASA enables the creation of a service object group that can contain a mix of TCP services, UDP services, ICMP-type services, and any protocol, such as ESP, GRE, and TCP.
- **ICMP-Type** - The ICMP protocol uses unique types to send control messages (RFC 792). The ICMP-type object group can group the necessary types required to meet an organization's security needs, such as to create an object group called ECHO to group echo and echo-reply.
- **Security** - A security group object group can be used in features that support Cisco TrustSec by including the group in an extended ACL, which in turn can be used in an access rule.

Configure Common Object Groups

To configure a network object group, use the **object-group network** *grp-name* global configuration mode command. After entering the command, add network objects to the network group using the **network-object** and **group-object** commands.

To configure an ICMP object group, use the **object-group icmp-type** *grp-name* global configuration mode command. After entering the command, add ICMP objects to the ICMP object group using the **icmp-object** and **group-object** commands.

The example displays a sample network object group configuration.

```
NETSEC-ASA(config)# object-group network ADMIN-HOST
NETSEC-ASA(config-network-object-group)# description Administrative hosts
NETSEC-ASA(config-network-object-group)# network-object host 192.168.1.3
NETSEC-ASA(config-network-object-group)# network-object host 192.168.1.4
NETSEC-ASA(config-network-object-group)# exit
NETSEC-ASA(config)# object-group network ALL-HOSTS
NETSEC-ASA(config-network-object-group)# description All inside hosts
NETSEC-ASA(config-network-object-group)# network-object 192.168.1.32 255.255.255.240
NETSEC-ASA(config-network-object-group)# group-object ADMIN-HOST
NETSEC-ASA(config-network-object-group)# exit
NETSEC-ASA(config)# show run object-group
object-group network ADMIN-HOST
 description Administrative host IP addresses
 network-object host 192.168.1.3
 network-object host 192.168.1.4
object-group network ALL-HOSTS
 network-object 192.168.1.32 255.255.255.240
 group-object ADMIN-HOST
NETSEC-ASA(config)#
```

21.4 ASA ACLs

ASA ACLs

These are the similarities between ASA ACLs and IOS ACLs:

- ACLs are made up of one or more ACEs. ACEs are applied to a protocol, a source and destination IP address, a network, or the source and destination ports.
- ACLs are processed sequentially from top down.
- A criteria match will cause the ACL to be exited.
- There is an implicit deny any at the bottom.
- Remarks can be added per ACE or ACL.
- Only one access list can be applied per interface, per protocol, per direction.
- ACLs can be enabled/disabled based on time ranges.

These the differences between ASA ACLs and IOS ACLs:

- The ASA uses a network mask (e.g., 255.255.255.0) and not a wildcard mask (e.g. 0.0.0.255).
- ACLs are always named instead of numbered.
- By default, interface security levels apply access control without an ACL configured.

Types of ASA ACL Filtering

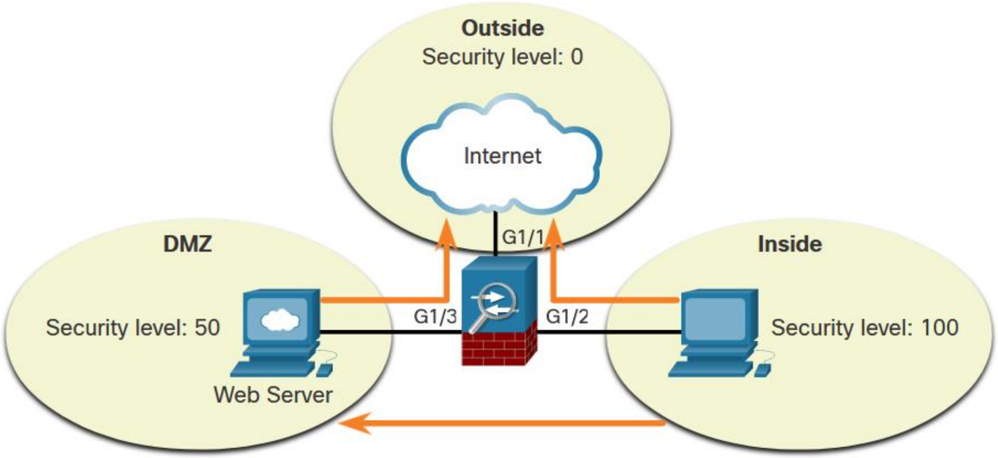
ACLs on a security appliance can be used not only to filter packets that are passing through the appliance but also to filter packets destined for the appliance.

- **Through-traffic filtering** - Traffic passing through the ASA from one interface to another interface. The configuration is completed in two steps: configure the ACL, then apply the ACL to an interface.
- **To-the-box-traffic filtering** - management access rule that applies to traffic that terminates at the ASA.

ASA devices differ from their router counterparts because of interface security levels. By default, security levels apply access control without an ACL configured.

Types of ASA ACL Filtering (Cont.)

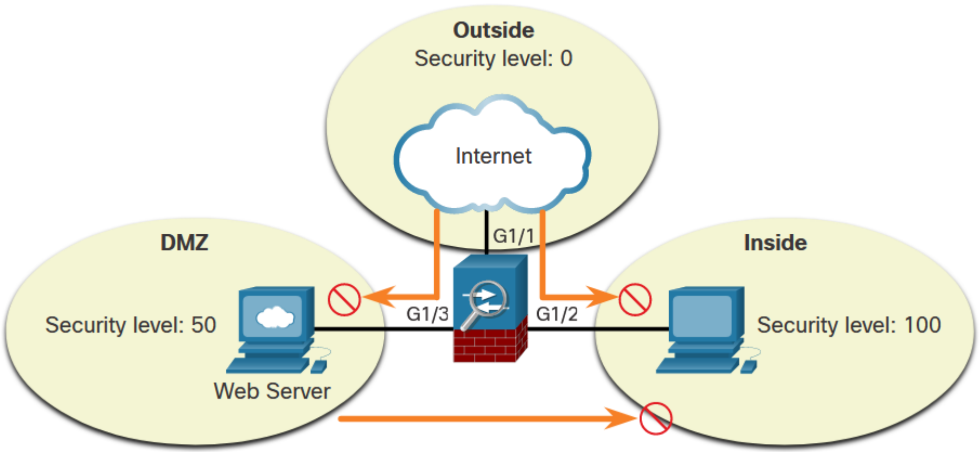
ASA devices differ from their router counterparts because of interface security levels. By default, security levels apply access control without an ACL configured.



More secure interfaces can access less secure interfaces.

Types of ASA ACL Filtering (Cont.)

However, a host from an outside interface with security level 0 cannot access the inside higher-level interface, as shown below.

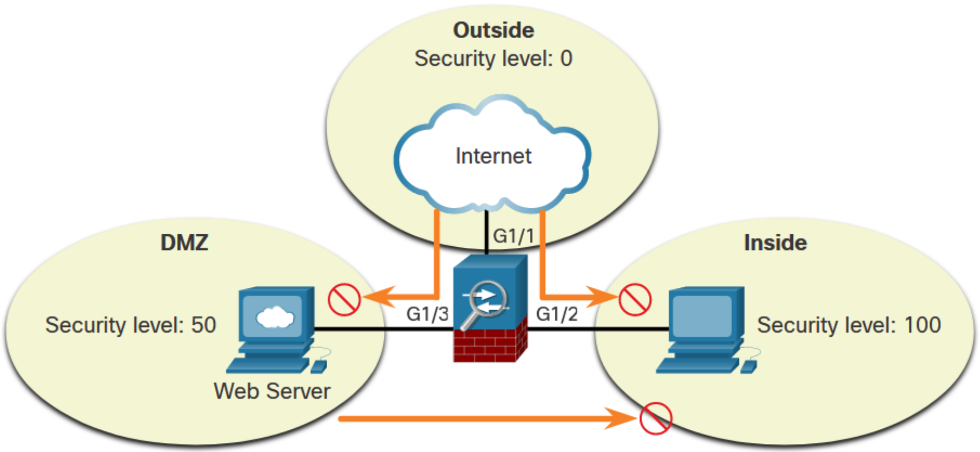


Less secure interfaces are blocked from accessing more secure interfaces.

Types of ASA ACL Filtering (Cont.)

A from an outside interface with security level 0 cannot access the inside higher-level interface, as shown in the figure.

Use the **same-security-traffic permit inter-interface** to enable traffic between interfaces with the same security level. Use the **same-security-traffic permit intra-interface** command to allow traffic to enter and exit the same interface



Less secure interfaces are blocked from accessing more secure interfaces.

Types of ASA ACLs

The ASA supports five types of access lists:

- **Extended access list** - The most common type of ACL.
- **Standard access list** - Unlike IOS where a standard ACL identifies the source host/network, ASA standard ACLs are used to identify the destination IP addresses. Standard access lists cannot be applied to interfaces to control traffic.
- **EtherType access list** - An EtherType ACL can be configured only if the security appliance is running in transparent mode.
- **Webtype access list** - Used for filtering for clientless SSL VPN traffic. These ACLs can deny access based on URLs or destination addresses.
- **IPv6 access list** - Used to determine which IPv6 traffic to block and which traffic to forward at router interfaces.

Use the **help access-list** privileged EXEC command to display the syntax for all of the ACLs supported on an ASA platform.

Types of ASA ACLs (Cont.)

The table provides examples of the uses of extended ACLs.

ACL Use	Description
Control network access for IP traffic	The ASA does not allow any traffic from a lower security interface to a higher security interface unless it is explicitly permitted by an extended access list.
Identify traffic for AAA rules	AAA rules use access lists to identify traffic.
Identify addresses for NAT	Policy NAT lets you identify local traffic for address translation by specifying the source and destination addresses in an extended access list.
Establish VPN access	Extended access list can be used in VPN commands.
Identify traffic for Modular Policy Framework (MPF)	• Access lists can be used to identify traffic in a class map, which is used for features that support MPF. • Features that support MPF include TCP, general connection settings, and inspection.

Types of ASA ACLs (Cont.)

The table provides examples of uses of standard ACLs.

ACL Use	Description
Identify OSPF destination network in route maps	Standard access lists include only the destination address. It can be used to control the redistribution of OSPF routes.
VPN filters	Filter traffic for LAN-to-LAN (L2L), Cisco VPN Client, and the Cisco AnyConnect Secure Mobility Client traffic.

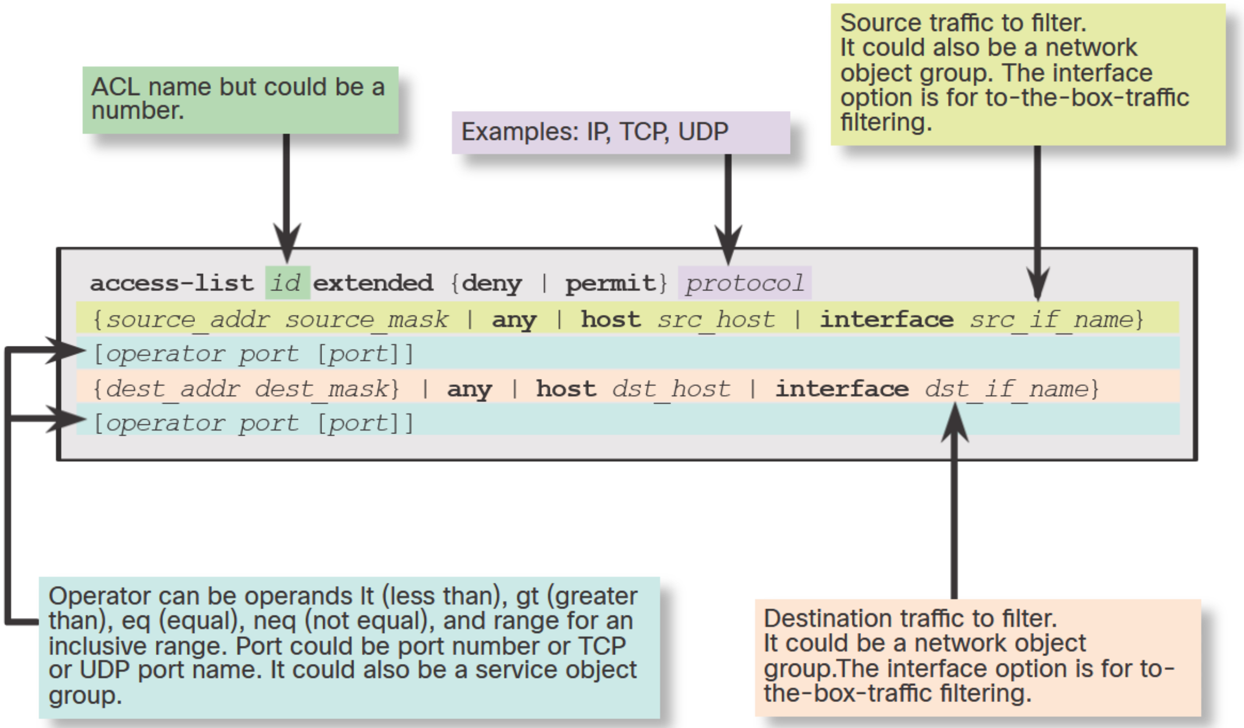
The table provides examples of uses of IPv6 ACLs.

ACL Use	Description
Control network access for IPv6 networks	Can be used to add and apply access lists to control traffic in IPv6 networks.

Syntax for Configuring an ASA ACL

IOS and ASA ACLs have similar elements, but some options vary with the ASA.

There are many options that can be used with ACLs. However, for most needs, a more useful and condensed version of the syntax is shown in the figure.



Syntax for Configuring an ASA ACL (Cont.)

The table describes elements of an ASA ACL.

Element	Description
ACL id	The name of the ACL.
Action	Can be **permit** or **deny**.
Protocol number - Source	Can be IP for all traffic, or the name / IP protocol number (0-250) including icmp (1), tcp (6), udp (17), or a protocol object-group.
Source	• Identifies the source and can be **any**, a host, a network, or a network object group. • For to-the-box-traffic filtering, the interface keyword is used to specify the source interface of the ASA.
Source port operator	• (Optional) Operand is used in conjunction with the source port. • Valid operands include **lt** (less than), **gt** (greater than), **eq** (equal), **neq** (not equal), and **range** for an inclusive range.
Source port	(Optional) Can be the actual TCP or UDP port number, select port names, or service object group.
Destination	• Identifies the destination and like the source, it can be **any**, a host, a network, or a network object group. • For to-the-box-traffic filtering, the interface keyword is used to specify the destination interface of the ASA.
Destination port operator	• (Optional) Operand is used in conjunction with the destination port. • Valid operands are the same as the source port operands.
Destination port	(Optional) Can be the actual TCP or UDP port number, select port names, or service object group.
Log	Can set elements for syslog including severity level and log interval.
Time range	(Optional) Specify a time range for the ACE.

Syntax for Applying an ASA ACL

The example displays the command syntax and parameter description for applying the ACL to an interface using the **access-group** command syntax. To verify ACLs, use the **show access-list** and **show running-config access-list** commands. To erase a configured ACL, use the **clear configure access-list** *id* command.

```
access-group id { in | out } interface if_name [ per-user-override | control-plane ]
```

Syntax	Description
`access-group`	Keyword used to apply an ACL to an interface.
id	The name of the actual ACL to be applied to an interface.
`in`	The ACL will filter inbound packets.
`out`	The ACL will filter outbound packets.
`interface`	Keyword to specify the interface to which to apply the ACL.
`if_name`	The name of the interface to which to apply an ACL.
`per-user-override`	Option that allows downloadable ACLs to override the entries on the interface ACL.
`control-plane`	Keyword to specify whether the applied ACL analyzes traffic destined to ASA for management purposes.

ASA ACL Examples

Example 1

- ACL allows all hosts on the inside network to go through the ASA.
- By default, all other traffic is denied unless explicitly permitted.

```
NETSEC-ASA(config)# access-list ACL-IN extended permit ip any any
NETSEC-ASA(config)# access group ACL-IN in interface INSIDE
```

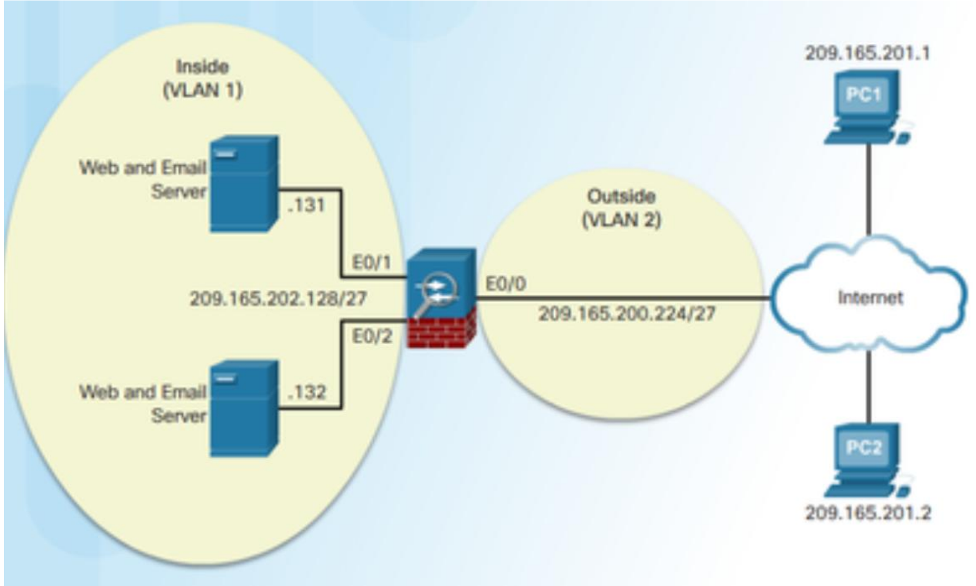
Example 2

- ACL prevents hosts on 192.168.1.0/24 from accessing the 209.165.201.0/27 network.
- Internal hosts are permitted access to all other addresses.
- All other traffic is implicitly denied.

```
NETSEC-ASA(config)# access-list ACL-IN extended deny ip 192.168.1.0 255.255.255.0 209.165.201.0
255.255.255.224
NETSEC-ASA(config)# access-list ACL-IN extended permit ip any any
NETSEC-ASA(config)# access-group ACL-IN in interface INSIDE
```

ACLs and Object Groups

Consider the sample topology in the figure in which access from two trusted, remote hosts, PC1 and PC2, should be allowed to the two internal for web and email servers. All other traffic attempting to pass through the ASA should be dropped and logged.



The ACL would require two ACEs for each PC to accomplish the task. The implicit **deny any** drops and logs any packets that do not match email or web services. As shown in the example, ACLs should always be thoroughly documented using the **remark** command.

To verify the ACL syntax, use the **show running-config access-list** and **show access-list** commands, as shown in the example.

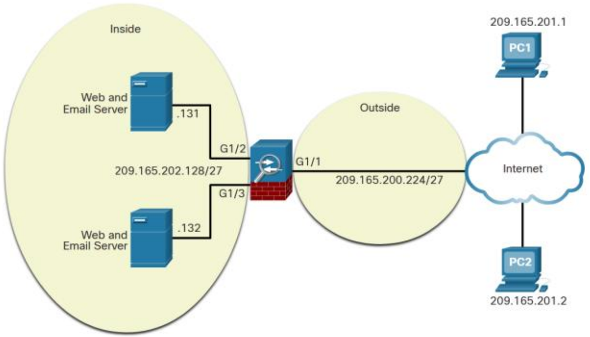
ACL Using Object Groups Examples

Object grouping is a way to group similar items together to reduce the number of ACEs.

Object grouping can cluster network objects into one group and outside hosts into another, as shown in the following syntax. The security appliance can also combine both TCP services into a service object group.

```
ciscoasa(config)# access-list id extended { deny | permit } protocol object-group
source_net-obj-grp_id object-group dest_net-obj-grp_id object-group service-obj-grp_id
```

ACL Using Object Groups Examples (Cont.)



This example shows how to configure the following three object groups:

- **NET-HOSTS** - Identifies two external hosts.
- **SERVERS** - Identifies servers providing email and web services.
- **HTTP-SMTP** - Identifies SMTP and HTTP protocols.

```
NETSEC-ASA(config)# object-group network NET-HOSTS
NETSEC-ASA(config-network-object-group)# description OG matches PC-A and PC-B
NETSEC-ASA(config-network-object-group)# network-object host 209.165.201.1
NETSEC-ASA(config-network-object-group)# network-object host 209.165.201.2
NETSEC-ASA(config-network-object-group)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# object-group network SERVERS
NETSEC-ASA(config-network-object-group)# description OG matches Web / Email Servers
NETSEC-ASA(config-network-object-group)# network-object host 209.165.202.131
NETSEC-ASA(config-network-object-group)# network-object host 209.165.202.132
NETSEC-ASA(config-network-object-group)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# object-group service HTTP-SMTP tcp
NETSEC-ASA(config-service-object-group)# description OG matches SMTP / WEB traffic
NETSEC-ASA(config-service-object-group)# port-object eq smtp
NETSEC-ASA(config-service-object-group)# port-object eq www
NETSEC-ASA(config-service-object-group)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# access-list ACL-IN remark Only permit PC-A / PC-B -> Internal Servers
NETSEC-ASA(config)# access-list ACL-IN extended permit tcp object-group NET-HOSTS object-
group SERVERS object-group HTTP-SMTP
```

21.5 NAT Services on an ASA

ASA NAT Overview

NAT can be deployed using one of the methods:

- **Inside NAT**
- **Outside NAT**
- **Bidirectional NAT**

Specifically, the Cisco ASA supports the following common types of NAT:

- **Dynamic PAT** - This is a many-to-one translation. This is also known as NAT with overload. Usually, an inside pool of private addresses overloading an outside interface or outside address.
- **Static NAT** - This is a one-to-one translation. Usually an outside address mapping to an internal server.
- **Policy NAT** - Policy-based NAT is based on a set of rules.
- **Identity NAT** - A real address is statically translated to itself, essentially bypassing NAT.

Configure Dynamic NAT

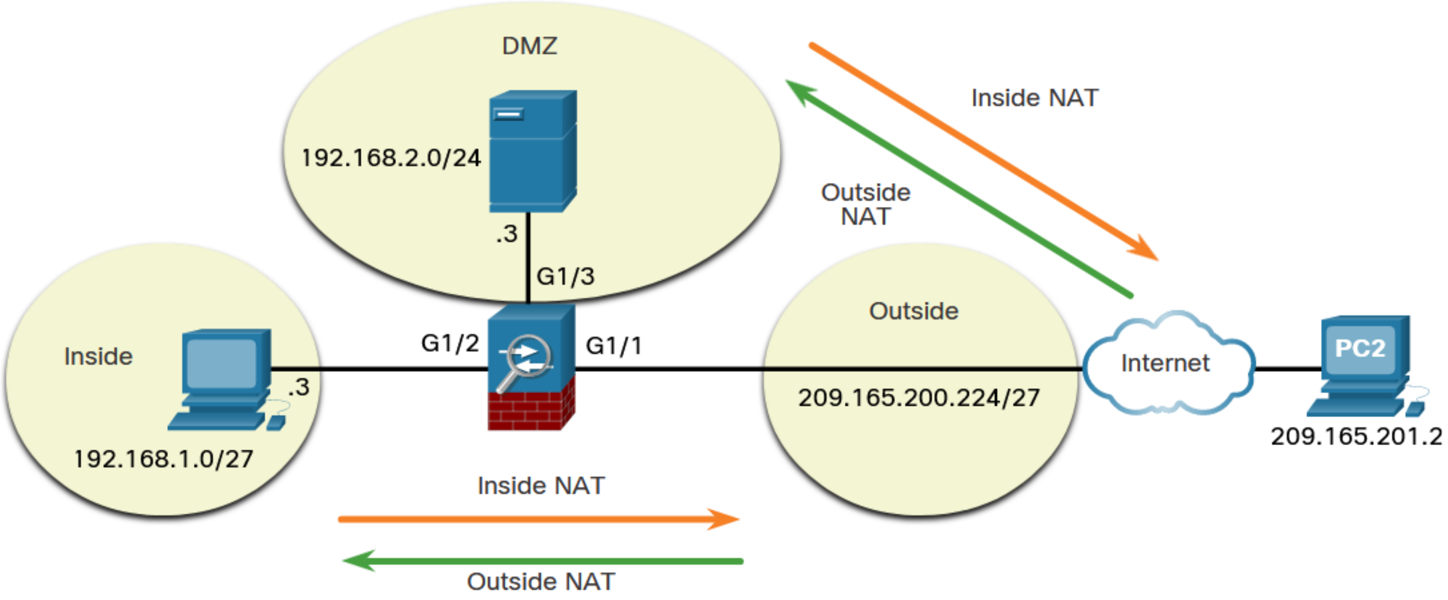
To configure network object dynamic NAT, two network objects are required:

- Identify the pool of public IP addresses with the **range** or **subnet** network object commands.
- Identify the internal addresses to be translated with the **range** or **subnet** network object commands.

The two network objects are then bound together using **nat** [(*real_if_name*,*mapped_if_name*)] **dynamic** *mapped_obj* [**interface** [**ipv6**]] [**dns**] network object command. The *real_if_name* is the prenat interface. The *mapped_if_name* is the postnat interface. Notice that there is no space after the comma in the command syntax.

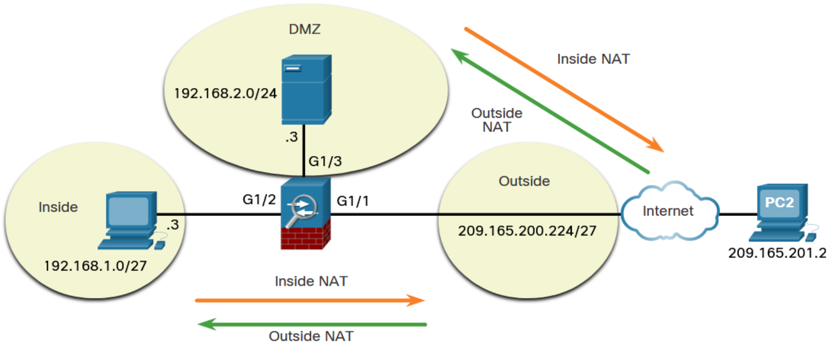
Configure Dynamic NAT (Cont.)

In this dynamic NAT example, the inside hosts on the 192.168.1.0/27 network will be dynamically assigned a range of public IP address from 209.165.200.240 to 209.165.200.248.



Configure Dynamic NAT (Cont.)

The example displays a sample dynamic NAT configuration to accomplish this task. The **PUBLIC** network object identifies the public IP addresses to be translated to while the **DYNAMIC-NAT** object identifies the internal addresses to be translated and is bound to the **PUBLIC** network object with the **nat** command.




```
NETSEC-ASA(config)# object network PUBLIC
NETSEC-ASA(config-network-object)# range 209.165.200.240 209.165.200.248
NETSEC-ASA(config-network-object)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# object network DYNAMIC-NAT
NETSEC-ASA(config-network-object)# subnet 192.168.1.0 255.255.255.224
NETSEC-ASA(config-network-object)# nat (INSIDE,OUTSIDE) dynamic PUBLIC
NETSEC-ASA(config-network-object)# end
NETSEC-ASA#
```

Configure Dynamic NAT (Cont.)

The following example shows how to allow inside hosts to ping outside hosts.

```
NETSEC-ASA(config)# policy-map global_policy
NETSEC-ASA(config-pmap)# class inspection_default
NETSEC-ASA(config-pmap-c)# access-list ICMPACL extended permit icmp any any
NETSEC-ASA(config)# access-group ICMPACL in interface OUTSIDE
NETSEC-ASA(config)#
```

To verify the network address translation, use **show xlate**, **show nat**, and **show nat detail** commands.

Configure Dynamic PAT

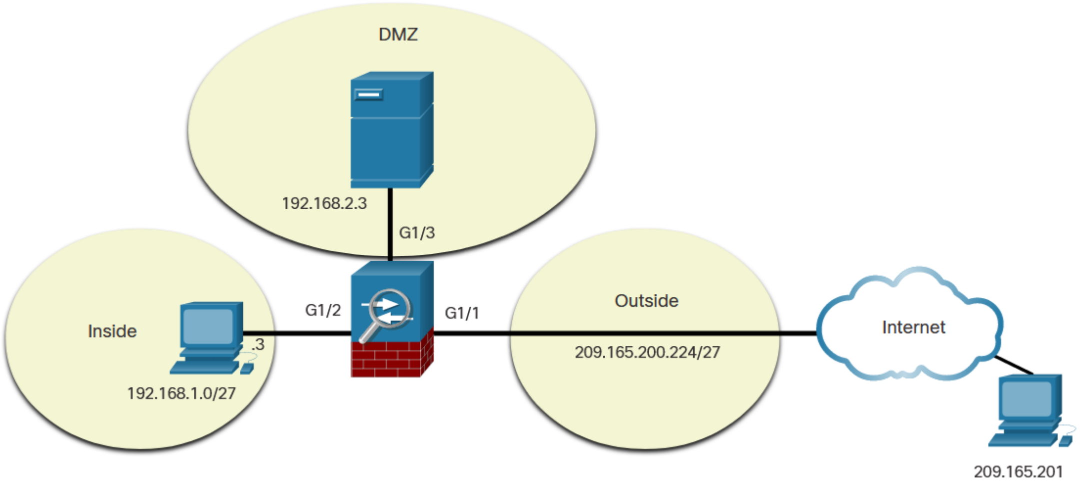
To enable inside hosts to overload the outside address, use **nat** [(*real_if_name*,*mapped_if_name*)] **dynamic interface** command, as shown in the example.

```
NETSEC-ASA(config)# object network INSIDE-NET
NETSEC-ASA(config-network-object)# subnet 192.168.1.0 255.255.255.224
NETSEC-ASA(config-network-object)# nat (inside,outside) dynamic interface
NETSEC-ASA(config-network-object)# end
NETSEC-ASA#
```

Configure Static NAT

Static NAT is configured when an inside address is mapped to an outside address. For instance, static NAT can be used when a server must be accessible from the outside.

To configure static NAT, use the **nat** [(*real_if_name*,*mapped_if_name*)] **static** m*apped-inline-host-ip* network object command. The example configuration is on the next slide.



Configure Static NAT (Cont.)

```
NETSEC-ASA(config)# object network DMZ-SERVER
NETSEC-ASA(config-network-object)# host 192.168.2.3
NETSEC-ASA(config-network-object)# nat (DMZ,OUTSIDE) static 209.165.200.227
NETSEC-ASA(config-network-object)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip any host 192.168.2.3
NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE
NETSEC-ASA(config)#
NETSEC-ASA(config)# policy-map global_policy
NETSEC-ASA(config-pmap)# class inspection_default
NETSEC-ASA(config-pmap-c)# access-list ICMPACL extended permit icmp any any
NETSEC-ASA(config)# access-group ICMPACL in interface DMZ
NETSEC-ASA(config)#
```

21.6 AAA

AAA Review

Authentication, authorization, and accounting (AAA) provides an extra level of protection and user control. Using AAA only, authenticated and authorized users can be permitted to connect through the ASA.

Authorization controls access, per user, after users are authenticated. Authorization controls the services and commands that are available to each authenticated user.

Accounting tracks traffic that passes through the ASA, enabling administrators to have a record of user activity.

Local Database and Servers

Use the **username** *name* **password** *password* [**privilege** *priv-level*] command to create local user accounts. To erase a user from the local database, use the **clear config username** [*name*] command. To view all user accounts, use the **show running-config username** command.

To configure a TACACS+ or RADIUS server, use the commands listed in the table.

ASA Command	Description
aaa-server *server-tag* **protocol** *protocol*	Creates a TACACS+ or RADIUS AAA server group.
aaa-server *server-tag* [(-interface name)] **host** {*server-ip* | *name* } [*key*]	Configures a AAA server as part of a AAA server group. Also configures AAA server parameters that are host-specific.

The example shows configuration of a AAA TACACS+ server on an ASA 5506-X.

```
NETSEC-ASA(config)# username Admin password class privilege 15
NETSEC-ASA(config)# show run username
username Admin password ***** pbkdf2 privilege 15
NETSEC-ASA(config)# aaa-server TACACS-SVR protocol tacacs+
NETSEC-ASA(config-aaa-server-group)# aaa-server TACACS-SVR (DMZ) host 192.168.2.3
NETSEC-ASA(config-aaa-server-host)# exit
NETSEC-ASA(config)# show run aaa-server
aaa-server TACACS-SVR protocol tacacs+
aaa-server TACACS-SVR (DMZ) host 192.168.2.3
NETSEC-ASA(config)#
```

AAA Configuration

To authenticate users who access the ASA CLI over a console (**serial**), SSH, HTTPS (ASDM), or Telnet connection, or to authenticate users who access privileged EXEC mode using the **enable** command, use the **aaa authentication enable console** command in global configuration mode.

```
ciscoasa(config)# aaa authentication { serial | enable | telnet | ssh | http } console { LOCAL
| server-group [ LOCAL ]}
```

AAA Configuration (Cont.)

The example provides a sample AAA configuration that is then verified and tested.

```
NETSEC-ASA(config)# aaa authentication serial console TACACS-SVR LOCAL
NETSEC-ASA(config)# aaa authentication ssh console TACACS-SVR LOCAL
NETSEC-ASA(config)# aaa authentication http console TACACS-SVR LOCAL
NETSEC-ASA(config)# aaa authentication telnet console TACACS-SVR LOCAL
NETSEC-ASA(config)# aaa authentication enable console TACACS-SVR LOCAL
NETSEC-ASA(config)#
NETSEC-ASA(config)# show run aaa
aaa authentication serial console TACACS-SVR LOCAL
aaa authentication ssh console TACACS-SVR LOCAL
aaa authentication http console TACACS-SVR LOCAL
aaa authentication telnet console TACACS-SVR LOCAL
aaa authentication enable console TACACS-SVR LOCAL
aaa authentication login-history
NETSEC-ASA(config)# exit
NETSEC-ASA# exit

Logoff

Username: Admin
Password: *****
-----------------------------------------------
Authorized access only!
You have logged into a secure device.
-----------------------------------------------

User Admin logged in to NETSEC-ASA
Logins over the last 2 days: 4.  Last login: 10:14:48 UTC Feb 11 2021 from console
Failed logins since the last login: 0.
Type help or '?' for a list of available commands.
NETSEC-ASA>
```

21.7 Service Policies on an ASA

Overview of MPF

A Modular Policy Framework (MPF) configuration defines a set of rules for applying firewall features, such as traffic inspection and QoS, to the traffic that traverses the ASA. MPF allows granular classification of traffic flows, which enables the application of different advanced policies to different flows.

Cisco MPF uses three configuration objects to define modular, object-oriented, hierarchical policies:

- Class Maps - What are we looking for?

```
ciscoasa(config)# class-map class-name
```

- Policy Maps - What shall we do with it?

```
ciscoasa(config)# policy-map policy-name
```

- Service Policy - Where do we do it?

```
ciscoasa(config)# service-policy serv-name [ global | interface if-name ]
```

Overview of MPF (Cont.)

There are four steps to configure MPF on an ASA:

Step 1. (Optional) Configure extended ACLs to identify granular traffic that can be specifically referenced in the class map.

Step 2. Configure the class map to identify traffic.

Step 3. Configure a policy map to apply actions to those class maps.

Step 4. Configure a service policy to attach the policy map to an interface.

Configure Class Maps

To create a class map and enter class-map configuration mode, use the **class-map** *class-map-name* global configuration mode command.

Next, traffic to match should be identified using the **match any** (matches all traffic) or **match access-list** *access-list-name* commands to match traffic specified by an extended access list.

```
NETSEC-ASA(config)# access-list UDP permit udp any any
NETSEC-ASA(config)# access-list TCP permit tcp any any
NETSEC-ASA(config)# access-list SERVER permit ip any host 10.1.1.1
NETSEC-ASA(config)#
NETSEC-ASA(config)# class-map ALL-TCP
NETSEC-ASA(config-cmap)# description This class-map matches all TCP traffic
NETSEC-ASA(config-cmap)# match access-list TCP
NETSEC-ASA(config-cmap)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# class-map ALL-UDP
NETSEC-ASA(config-cmap)# description This class-map matches all UDP traffic
NETSEC-ASA(config-cmap)# match access-list UDP
NETSEC-ASA(config-cmap)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# class-map ALL-HTTP
NETSEC-ASA(config-cmap)# description This class-map matches all HTTP traffic
NETSEC-ASA(config-cmap)# match port TCP eq http
NETSEC-ASA(config-cmap)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# class-map TO-SERVER
NETSEC-ASA(config-cmap)# description Class map matches traffic  10.1.1.1
NETSEC-ASA(config-cmap)# match access-list SERVER
NETSEC-ASA(config-cmap)# exit
NETSEC-ASA(config)#
```

Define and Activate a Policy

Use the **policy-map** *policy-map-name* global configuration mode command, to apply actions to the Layer 3 and 4 traffic.

In policy-map configuration mode, config-pmap, use the following commands:

- **description** - Add description text.
- **class** *class-map-name* - Identify a specific class map on which to perform actions.

These are the three most common commands available in policy map configuration mode:

- **set connection** - Sets connection values.
- **inspect** - Provides protocol inspection servers.
- **police** - Sets rate limits for traffic in this class.

To activate a policy map globally on all interfaces or on a targeted interface, use the **service-policy** *policy-map-name* [**global** | **interface** *intf*] global configuration mode command to enable a set of policies on an interface.

Define and Activate a Policy (Cont.)

The example configures the policy map. Its associated service policy is applied globally.

```
NETSEC-ASA(config)# access-list TFTP-TRAFFIC permit udp any any eq 69
NETSEC-ASA(config)#
NETSEC-ASA(config)# class-map CLASS-TFTP
NETSEC-ASA(config-cmap)# match access-list TFTP-TRAFFIC
NETSEC-ASA(config-cmap)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# policy-map POLICY-TFTP
NETSEC-ASA(config-pmap)# class CLASS-TFTP
NETSEC-ASA(config-pmap-c)# inspect tftp
NETSEC-ASA(config-pmap-c)# exit
NETSEC-ASA(config-pmap)# exit
NETSEC-ASA(config)#
NETSEC-ASA(config)# service-policy POLICY-TFTP global
NETSEC-ASA(config)#
```

Packet Tracer - Configure ASA Basic Settings and Firewall Using the CLI

In this comprehensive Packet Tracer activity, you will complete the following objectives:

- Verify connectivity and explore the ASA.
- Configure basic ASA settings and interface security levels using the CLI.
- Configure routing, address translation, and inspection policy using the CLI.
- Configure DHCP, AAA, and SSH.
- Configure a DMZ, Static NAT, and ACLs.

Optional Lab - Configure ASA Network Services, Routing, and DMZ with ACLs Using CLI

In this comprehensive lab, you will complete the following objectives:

- Part 1: Configure Basic Device Settings
- Part 2: Configure Routing, Address Translation, and Inspection Policy Using the CLI
- Part 3: Configure DHCP, AAA, and SSH
- Part 4: Configure DMZ, Static NAT, and ACLs

21.8 ASA Firewall Configuration Summary

What Did I Learn in this Module?

- The ASA CLI contains command prompts similar to that of a Cisco IOS router.
- Many commands are similar to those in other versions of IOS, however many differences also exist.
- The ASA 5506-X with FirePOWER Services ships with a default configuration that, in most instances, is sufficient for a basic SOHO deployment.
- The ASA 5506-X has eight Gigabit Ethernet interfaces that can be configured to carry traffic on different Layer 3 networks. The G1/1 interface is frequently configured as the outside interface to the ISP.
- Basic configuration of interfaces includes IP addressing, naming, and setting the security level.
- If the interface is configured with DHCP, a default route from an upstream device can automatically be configured on the ASA. Otherwise, a default route must be manually configured.
- Objects make it easy to maintain configurations because an object can be modified in one place and the change will be reflected in all other places that are referencing it.
- Network objects can include host addresses, subnets, ranges of addresses, and FQDNs.
- Service objects can refer to different network services and protocols.
- Object groups are collections of objects that are related.
- Network object groups can also be used in configurations including ACLs and NAT.

What Did I Learn in this Module? (Cont.)

- ASA ACLs differ from IOS ACLs in that they use a network mask (e.g., 255.255.255.0) instead of a wildcard mask (e.g. 0.0.0.255).
- ASA ACLs must be grouped with an interface in order to go into effect. Object groups can be used with ASA ACLs to limit the number of ACEs that are required in a list.
- There are three NAT deployment methods for the ASA: inside NAT, outside NAT, and bidirectional NAT.
- The ASA supports four types of NAT: dynamic NAT with overload, static NAT, policy NAT, and identity NAT.
- Cisco ASAs can be configured to authenticate access using a local user database or an external server for authentication or both.
- A Modular Policy Framework (MPF) configuration defines a set of rules for applying firewall features, such as traffic inspection and QoS, to the traffic that traverses the ASA.
- Class maps are used to identify the traffic that will be processed by MPF.
- Policy maps define what will be done to the identified traffic.
- Service policies identify which interfaces the policy map should be applied to.

New Terms and Commands

- **configure factory-default**
- **domain-name name**
- **key config-key password-encryption** [*new-pass* [*old-pass*]]
- **password encryption aes**
- **show password encryption**
- **ip address dhcp setroute**
- **ip address pppoe**
- **ip address pppoe setroute**
- **nameif** *if_name*
- **security-level** *value*
- **route** *interface-name* **0.0.0.0 0.0.0.0** *next-hop-ip-address*
- {**passwd** | **password**} *password*
- **telnet** { *ipv4_address mask* | *ipv6_address***/***prefix* } *if_name*
- **telnet timeout** *minutes*
- **aaa authentication telnet console LOCAL**
- **clear configure telnet**
- **ssh** { *ip_address mask* | *ipv6_address***/***prefix* } *if_name*
- **ssh version** *version_number*
- **ssh timeout** *minutes*
- **clear configure ssh**
- Network Time Protocol (NTP)
- **ntp authenticate**
- **ntp trusted-key** *key_id*
- **ntp authentication-key** *key_id* **md5** *key*
- **ntp server** *ip_address* [**key** *key_id*]

New Terms and Commands

- **dhcpd address** *IP_address1* [*-IP_address2*] *if_name*
- **dhcpd dns** *dns1* [*dns2*]
- **dhcpd lease** *lease_length*
- **dhcpd domain** *domain_name*
- **dhcpd enable** *if_name*
- **object groups**
- **object network** *object-name*
- **object service** *object-name*
- **show run object service**
- **object-group network** *grp-name*
- **object-group icmp-type** *grp-name*
- Through-traffic filtering
- To-the-box-traffic filtering
- **same-security-traffic permit inter-interface**
- **same-security-traffic permit intra-interface**
- **access-group id** { **in** | **out** } **interface** *if_name* [**per-user-override** | **control-plane**]
- **access-list id extended** { **deny** | **permit** } *protocol* object-**group** *source_net-obj-grp_id* **object-group** *dest_net-obj-grp_id* **object-group** *service-obj-grp_id*
- *nat* [(*real_if_name*,*mapped_if_name*)] **dynamic** *mapped_obj* [**interface** [**ipv6**]] [**dns**]
- **show xlate**
- **show nat**
- **show nat detail**
- **nat** [(*real_if_name*,*mapped_if_name*)] static mapped-inline-host-ip
- **aaa-server** *server-tag* **protocol** *destined*

New Terms and Commands

- aaa-server *server-tag* [(**(***if_name***)**] **host** {*server-ip* | *name* } [*key*]
- **username** *name* **password** *password* [**privilege** *priv-level*]
- **clear config username** [*name*]
- Modular Policy Framework (MPF)
- **class-map** *class-name*
- **policy-map** *policy-name*
- **service-policy** *serv-name* [**global** | **interface** *if-name*]